

Lasai Barreñada

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PROFESSIONAL EXPERIENCE

KU Leuven

PhD Researcher in clinical prediction models (Leuven)

10/2022-Current

PhD research focused on clinical prediction models, including their development, validation, and implementation in clinical practice. Supervised by **Prof. Ben Van Calster**, **Prof. Laure Wynants**, and **Prof. Dirk Timmerman**.

Memorial Sloan Kettering Cancer Center

Visiting PhD Researcher (New York)

02/2025-06/2025

Four months research visit under supervision of **Prof. Andrew Vickers**, focused on evaluating prediction model's clinical utility.

Julius center - UMC Utrecht

Visiting PhD Researcher (Utrecht)

10/2024-02/2025

Four months research visit under supervision of **Prof. Maarten Van Smeden** and **Prof. Ewout Steyerberg**, focused on calibration performance of prediction models.

Basque Center for Applied Mathematics (BCAM)

Research Technician in Machine Learning (Bilbao)

12/2021-09/2022

Model development for COVID-19 prognosis under supervision of **Prof. Santiago Mazuelas**.

Spanish National Statistical Office (INE Spain)

Postgraduate Internship (Madrid)

10/2020-09/2021

EDUCATION

KU Leuven, PhD in Biomedical Sciences

2022-2026 (expected)

Complutense University of Madrid (UCM), European Master in Official Statistics (EMOS)

2019-2021

University of Granada, B.A. Marketing

2015-2019

PUBLICATIONS

1. **Barreñada, L.**, Campo, B. D. C., Wynants, L. & Calster, B. V. [Clustered flexible calibration plots for binary outcomes using random effects modeling](#). *Res Synth Methods* **In press**, 1–22 (2025).
2. Van Calster, B. *et al.* [Evaluation of performance measures in predictive artificial intelligence models to support medical decisions: Overview and guidance](#). *The Lancet Digital Health* **In press**, (2025).
3. Moro, F. *et al.* [External validation of ultrasound-based models for differentiating between benign and malignant adnexal masses: A nationwide prospective multicenter study \(IOTA phase 6\)](#). *American Journal of Obstetrics and Gynecology* **233**, 629.e1–629.e13 (2025).
4. **Barreñada, L.** *et al.* [Head-to-head comparison of the RMI and ADNEX models to estimate the risk of ovarian malignancy: A systematic review and meta-analysis of external validation studies](#). *BMJ Open* **15**, e04141 (2025).
5. Ramirez Zegarra, R. *et al.* [Standardization of the sonographic measurement of the head to perineum distance: A pilot study comparing techniques with and without perineal compression](#). *European Journal of Obstetrics & Gynecology and Reproductive Biology* **312**, 114520 (2025).
6. **Barreñada, L.** *et al.* [The fundamental problem of risk prediction for individuals: Health AI, uncertainty, and personalized medicine](#). vol. Preprint (2025).

7. Geysels, A. *et al.* [Artificial intelligence applied to ultrasound diagnosis of pelvic gynecological tumors: A systematic review and meta-analysis](#). *Gynecologic and Obstetric Investigation* **91**, 115–136 (2025).
8. Martínez-García, M., García-Gutierrez, S., **Barreñada, L.**, Inza, I. & Lozano, J. A. [Extending the learning using privileged information paradigm to logistic regression](#). *Neurocomputing* **615**, 128869 (2025).
9. **Barreñada, L.**, Dhiman, P., Timmerman, D., Boulesteix, A.-L. & Van Calster, B. [Understanding overfitting in random forest for probability estimation: A visualization and simulation study](#). *Diagnostic and Prognostic Research* **8**, 14 (2024).
10. **Barreñada, L. et al.** [ADNEX risk prediction model for diagnosis of ovarian cancer: Systematic review and meta-analysis of external validation studies](#). *bmjmed* **3**, e000817 (2024).
11. **Barreñada, L.**, Gálvez Sainz de Cueto, J. C. & Fernández Calatrava, J. [Timeliness reduction on industrial turnover index based on machine learning algorithms](#). *Statistical Journal of the LAOS* **38**, 1195–1205 (2022).

PEER REVIEW

Conducted various peer reviews for **Statistics in Medicine**, **Statistical methods in medical research**, **Nature Communications medicine**, **BMC medical informatics and decision making**, **Medical Decision Making**, **Diagnostic and prognostic research**, **Ultrasound in Obstetrics and Gynecology**, **International Journal of Gynecological Cancer**, **npj precision oncology** among others

CONFERENCE PRESENTATIONS

Problem of Individual Risk Estimation for Health AI (Talk) , SEB, Bilbao	01/2026
Updating the ADNEX model for surgically and conservatively managed patients (Talk) , ISUOG, Cancun	09/2025
Combining calibration plots (Talk) , MEMTAB, Birmingham	04/2025
Problem of Individual Risk Estimation for Health AI (Talk) , ENAR, New Orleans	03/2025
Head-to-head comparisons of RMI and ADNEX models (Talk) , ISUOG, Budapest	09/2024
Multicenter flexible calibration curves using random effects (Poster) , ISCB, Thessaloniki	07/2024
Drawing multicenter calibration curves (Talk) , SEB, Barcelona	02/2024
ADNEX model systematic review and meta-analysis (Talk) , ISUOG, Seoul	10/2023
Do we understand random forests? (Poster) , ISCB, Milan	08/2023
Do we understand random forests? (Poster) , YSM, Leicester	07/2023

AWARDS & PRIZES

Young Investigator Award in Gynaecology , issued by ISUOG	2023
Young Statisticians Prize (YSP) , 2 nd , issued by IAOS	2022

PROFESSIONAL MEMBERSHIPS

STRATOS early career adjunct member
 International Society For Clinical Biostatistics (ISCB) - Member of the early career subcommittee (ECB)
 International Biometric Society (IBS)
 International Society of Ultrasound in Obstetrics and Gynecology (ISUOG)
 International Ovarian Tumour Analysis group (IOTA)

LANGUAGES

Spanish: Native
Basque: Native
English: Full Working Proficiency
French: Full Working Proficiency